

TrES-1b

R-1848

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130731 · created 2026-08-03T07:58:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456504.7357 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.05
Transit depth (Rp/Rs)^2	3.01 +/- 1.73 [%]
Orbital Inclination (inc)	85.1 +/- 1.45
Airmass coefficient 1 (a1)	0.853 +/- 0.021
Airmass coefficient 2 (a2)	0.151 +/- 0.093
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	#2 - [321, 49]
Optimal Aperture	4.19
Optimal Annulus	6.49
Transit Duration (day)	0.056 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.05 vs 0.1358 (deviation 0.43 σ)
Duration fit vs published	0.056 d vs 0.1042 d (deviation 0.9 σ)
Mid-time O-C	2.6 min
Depth SNR	2.0
Residual scatter	2.46 %

Light curve

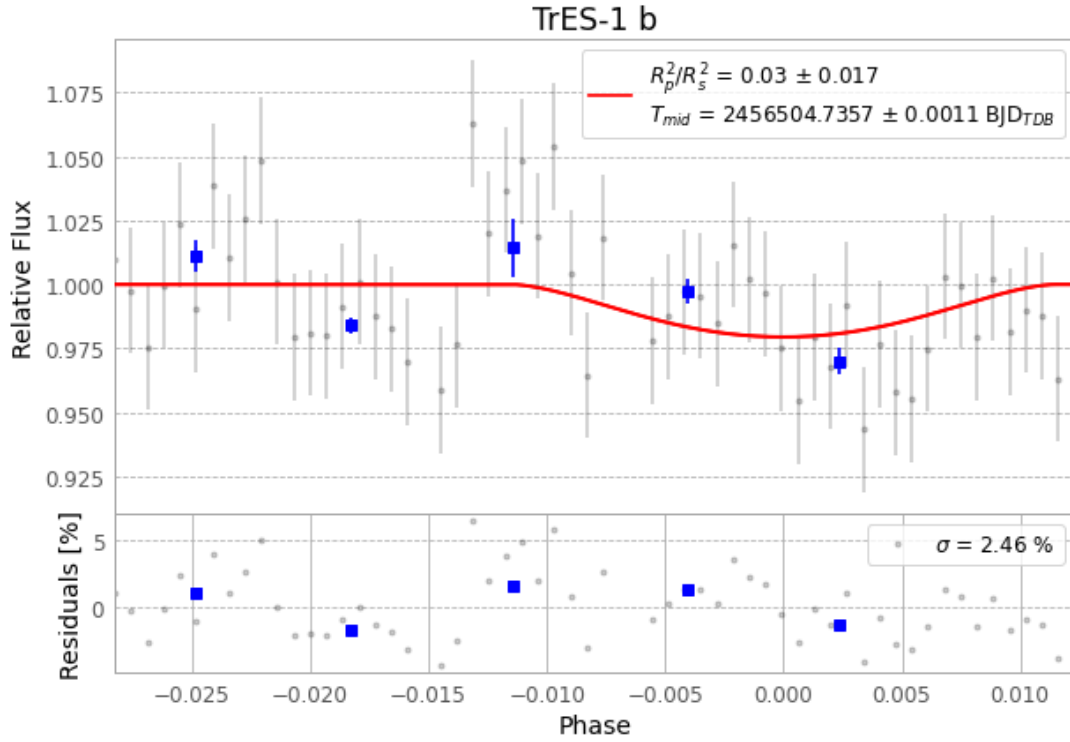


Figure 1 — FinalLightCurve_TrES-1 b_2013-07-30.png (SHA-256 fc850386a17efdd5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fa34e2913f3ec700...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), TRES-1130731033012.FITS → TRES-1130731062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130731.log (021bd50c0d69...), FinalLightCurve_TrES-1 b_2013-07-30.png (fc850386a17e...), FinalLightCurve_TrES-1 b_2013-07-30.csv (115e5b95161f...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:58Z.